ARCHIVING AND BACKUPS

Nesva

A desktop application that gets your files and your mail into the cloud — readable and browsable, or encrypted and versioned.

Version: 0.6.6
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Repository: github.com/Linux-Alex/Nesva
Licence: Apache-2.0
Platforms: Windows x86_64 · Linux x86_64

C++20 + Qt 6 Widgets | Rclone | Restic | SQLite | No cloud middleman

Pitch and user guide — English edition.

ENGLISH

What Nesva is, and who it is for

Nesva is a desktop application for two related but different jobs: **archiving** files into cloud storage, and **backing them up** with versions. It keeps those apart on purpose, because they are two different promises and it is dishonest to blur them.

Archive (a copy)
Files are copied into storage (iCloud, Google Drive, pCloud, B2, a NAS) and stay readable there. You can open them in the provider web interface or on a phone, without Nesva and without any special tool.

Backup (versions)
Restic writes encrypted snapshots. Every snapshot is a complete state you can restore at any time, including after an accidental deletion or ransomware. A retention policy decides what is kept.

Why bother

- **Two kinds of storage, one tool.** For "I have a copy in iCloud" and for "I have a restorable version from last week".
- **Mail as an archive.** Microsoft 365 mail is exported as `.eml` files: readable, portable, independent of a licence.
- **No secrets in commands or logs.** Passwords and tokens live in the system credential store, and logs are redacted before they are written.
- **No surprises.** Nesva never deletes a source, never syncs in two directions, and never writes into a destination that belongs to another job.
- **It runs without a window.** Schedules are executed by a background agent, even with the application closed.

> **The principle behind the code:** Nesva is the carrier, never the owner. Your files stay yours, in the cloud in a form the provider understands, and in the original place they were found.

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1. Installation

Windows

Download `nesva-0.6.6-windows-x86_64-setup.exe` from GitHub Releases and run it. It installs into `%LOCALAPPDATA%\Programs\Nesva`, per user, **with no administrator rights** and no UAC prompt. A Start-menu shortcut is created and an entry appears in Apps and features.

If you prefer it portable, take `nesva-0.6.6-windows-x86_64.zip`, unpack it anywhere and run `nesva.exe`. Qt and the compiler runtime are included, so nothing else has to be installed.

Linux

```
# Debian / Ubuntu - recommends rclone and restic
sudo apt install ./nesva_0.6.6_amd64.deb

# portable tree (Qt bundled, installs nothing into the system)
tar -xzf nesva-0.6.6-linux-x86_64.tar.gz
./nesva-0.6.6-linux-x86_64/bin/nesva
```

The package puts the whole application into its own directory, `/usr/lib/nesva`, with only symlinks in `/usr/bin`. That is why Nesva cannot disturb system Qt applications, which matters on KDE and Wayland.

> **Honest note:** code signing does not exist yet. Windows SmartScreen will warn on the first launch; that is expected and cannot be silenced without a certificate. macOS is not supported because it was never tested.

2. First run and the window

On first launch Nesva creates its database (`%LOCALAPPDATA%\nesva` or `~/.local/share/nesva`) and opens the **Dashboard**. Navigation is on the left, the activity dock at the bottom, and the status bar at the top shows the agent indicator.

Page	What it is for
Dashboard	Storage state, recent jobs, sizes, warnings.
Backup Jobs	Job list with *New job*, *Edit*, *Run now*, *Cancel*, *History*.
Storage	Local folders and remotes; add, test the connection, credential state.
Archive Explorer	What the archive holds: paths, sizes, SHA-256, verification state.
Mail	Microsoft 365 accounts, folder counts, *Estimate*, *Sync*.
Snapshots	Restic snapshots, repository check, retention policy.
Restore	Restoring from an archive or a snapshot, with an overwrite policy.
Activity	Timeline of jobs and application events.
Settings (Ctrl+8)	Appearance, logging, safety, locations, window behaviour, Microsoft 365, external tools, background agent, updates.

3. External tools: Rclone and Restic

Nesva ships neither program. It installs them with you watching: download URL, checksum, unpack command, target path and any elevation are shown before anything happens.

Settings → External tools → *Install Rclone…* and *Install Restic…*. The installer fetches the official release over HTTPS, verifies the published SHA-256, then unpacks and copies a single binary into your own directory, with no password.

```
nesva-cli tools restic        # preview only
nesva-cli tools restic --install
nesva-cli tools rclone --install --system  # all users, asks for pkexec
```

4. Storage: a local folder and clouds

Storage knows two kinds of destination: a local folder (external disk, NAS, another partition) and a remote handled by Rclone. A remote is described by the *remote name Rclone already knows* plus a base directory inside it.

Field	Meaning
Remote name	For example `icloud`, `b2kraken`, `gdrive`, `pcloud`.
Base directory	For example `Nesva/Archive`. Per-job subdirectories are created inside it.
Display name	How you refer to the storage in jobs.

Configuring the remote is built in: **Storage → Configure / re-authenticate…** opens a provider walkthrough (iCloud Drive, Google Drive, pCloud, Backblaze B2, OneDrive, WebDAV/NAS) on the left and a live `rclone config` session on the right. The Apple ID password and the two-factor code go straight into Rclone; Nesva does not store them.

> **Important:** before every job Nesva probes the destination. If the session expired it marks the storage as *authentication required* and stops the job with a clear message instead of failing file by file.

5. An archive job (files and folders)

Backup Jobs → New job and fill in the form:

Field	Value / meaning
Name	Anything that will still mean something to you in a year.
Mode	**Archive - copy files, no versions**.
Content	**Files and folders**.
Sources	One or more folders (Desktop, Documents, Downloads).
Destination	A storage from the list plus an optional subdirectory.
Exclusions	One pattern per line: `*.tmp`, `build/**`, `Thumbs.db`.
Verify checksums	Re-reads each uploaded file and compares it.
Bandwidth limit	KiB/s, so a job does not eat the whole link.
Repeat	Daily, weekly or monthly, with a time and a time zone.

The first run transfers everything; every later run is **incremental**: Nesva compares against the list of files already uploaded and sends only new or changed ones. With Rclone this happens as one transfer per source directory, which is several times faster than one transfer per file.

> **What Nesva refuses:** a destination that already holds another job output; a source equal to the destination or inside it; a destination without free space (it warns and suggests another path or provider).

6. Versioned backups

Same form, but **Mode: Backup**. Restic fields appear:

- **Repository** - a folder or remote where Restic lives (for example `icloud:Nesva/Restic`).
- **Password** - stored in the system credential store (Windows Credential Manager, Linux Secret Service). Restic receives it through a temporary file that is deleted immediately; it never appears on a command line.
- **Retention** - for example 7 daily, 4 weekly, 12 monthly. You can *preview* the effect before applying it.
- **Check after backup** - runs `restic check`.

```
nesva-cli repository init 3            # store password, restic init
nesva-cli snapshots list 3
nesva-cli retention preview 3         # what the policy would remove
nesva-cli retention apply 3 --prune
nesva-cli repository check 3 --read-data 5%
```

7. Microsoft 365: archiving mail

Mail is treated as another source, not as an afterthought. Register the application in Entra ID as a *public client* (no secret), with the redirect `http://localhost`, delegated permissions **Mail.Read** and **User.Read**, and admin consent.

Settings → Microsoft 365: fill in the tenant id, the client id and the folder for `.eml` files. Then **Connect…**, sign in through the browser, and choose folders on the **Mail** page.

```
nesva-cli mail connect --account you@example.com --tenant TENANT --client CLIENT
nesva-cli mail folders you@example.com
nesva-cli mail estimate you@example.com      # count and time estimate
nesva-cli mail sync you@example.com --to ~/Archive/Mail
```

Synchronisation is incremental (Graph delta): each run transfers only new mail. During a run you see a percentage, a rate and a remaining time. For a cloud archive, create a second job with **Content: Microsoft 365 mailbox** and a cloud destination.

8. Schedules, the agent and autostart

Schedules are executed by `nesva-agent`, a separate windowless process. Jobs therefore run with the window closed, and they survive a bad night: missed slots are caught up, failures retry with a growing backoff, and a maximum run duration is enforced.

Settings → Background agent offers *Start agent*, *Stop agent*, *Install autostart* and *Remove autostart*.

- **Windows:** a value in `HKCU\Software\Microsoft\Windows\CurrentVersion\Run` - no PowerShell, no scheduled task, no elevation, so it works under group policies that block scripting.
- **Linux:** a systemd user unit (`nesva-agent.service`) that runs inside your session and therefore sees your credentials.
- **Tray:** the window itself can start at sign-in into the notification area (`nesva --tray`), separately from the agent.

```
nesva-cli agent status
nesva-cli agent run 1        # ask the agent to start a job
nesva-cli agent stop
nesva-cli schedule list
```

9. Restoring and browsing an archive

Archive Explorer shows what a job contains: path, size, SHA-256 and whether the file was verified after upload. **Restore** writes a selection into a target folder, with a policy for files that already exist (skip, overwrite, rename).

```
nesva-cli archive list 1
nesva-cli archive restore 1 --to /media/disk/restore --policy skip
nesva-cli verify 1                          # re-check published files

# from an encrypted backup
nesva-cli backup restore 3 --snapshot latest --to /media/disk/restore
```

An archive can also be restored without Nesva: files sit in storage in their normal form, and mail as `.eml`, which opens in any mail client.

10. Maintenance and verification

```
nesva-cli doctor            # environment, database, space, tools, user context
nesva-cli status            # storage state and recent jobs
nesva-cli history list --limit 10
nesva-cli repository check 3 --read-data 5%
nesva-cli repository unlock 3              # remove stale locks
nesva-cli maintenance                      # cleanup and summary
```

Recommended rhythm: `verify` once a month for an archive, `repository check` once a month for a backup, and a trial restore of a few files at least twice a year. A backup that was never restored is only a hope.

11. Updates

Nesva updates itself from GitHub Releases, verifying the checksum before installing.

- **On start** (on by default): one check per six hours; a tray notification appears when a newer release exists.
- **Automatic, weekly** (on by default): on Sunday at **18:00 Europe/Ljubljana** the agent downloads, verifies, stops itself and the window, and runs the installer silently.
- **Manually: Settings → Updates →** *Check now* and *Install update*.
- Pre-releases and `-candidate` tags are never installed automatically.

```
nesva-cli update               # what is available
nesva-cli update --install     # download, verify, stop the agent, install
```

12. Security and privacy

Area	How it is handled
Secrets	The Microsoft 365 refresh token and the Restic password live in the system credential store; they are not in configuration files, logs or command lines.
Logs	Everything passes a redactor before it is written: registered secrets, `key=value`, Bearer/Basic headers, PEM private keys, JWT-shaped tokens.
Privileges	Everything runs as the signed-in user. Started by a service account (SYSTEM, LocalService, NetworkService, root) a job is refused, because it would archive the wrong profile.
Commands	No shell: every external program runs with the exact argument vector shown in the preview.
Sources	Never deleted, moved or renamed. Files that exist only as cloud placeholders are skipped so Nesva does not silently download them.
Network	No telemetry. The only automatic network activity is the update check and whatever synchronization you configured.

13. Command line (CLI) overview

`nesva-cli` covers everything the window does, plus automation. `--json` gives machine-readable output and the exit codes are stable (0 success, 1 usage error, 2 runtime failure, 3 not found, 4 not implemented).

Command	Purpose
`status`, `doctor`	State overview and environment diagnostics.
`jobs list | run | cancel`	List jobs, run one now (also `--dry-run`), cancel.
`storage list | test`	Storages and a connection test.
`archive list | restore`	Archive contents and restore.
`verify`, `history list`	Re-check published files, run history.
`mail connect | accounts | folders | estimate | sync | status`	Microsoft 365 from first sign-in to export.
`repository init | info | check | locks | unlock`	Restic repository.
`snapshots list`, `backup restore`	Snapshots and restore.
`retention preview | apply`	Retention: preview first, then apply.
`schedule list`, `agent status | run | cancel | reload | stop`	Schedules and control of the background agent.
`tools rclone | restic [--install]`	Preview and installation of the external tools.
`update [--install]`	Check for and install an update.

```
# a typical weekly scenario in a single scheduled task
nesva-cli jobs run 1 --quiet && nesva-cli jobs run 2 --quiet && nesva-cli verify 1
```

14. Troubleshooting

Symptom	What to do
Storage marked *authentication required*	Storage → Configure / re-authenticate; for iCloud enter a fresh two-factor code.
No space left	Nesva stops before writing; free space or choose another path or provider.
Files skipped as *cloud placeholder*	In your file manager choose *Always keep on this device*, then run the job again.
The agent is not running after a reboot	Settings → Background agent → *Install autostart*, then *Start agent*.
What happened last night	The **Activity** page or `nesva-cli history list`; the log file lives in the `logs` directory.
Unsure about the environment	`nesva-cli doctor` reports what is missing and which user Nesva runs as.

> **What Nesva does not do:** it does not sync in two directions, does not delete sources, does not offer web access to your files, does not encrypt an archive (encryption is what Restic

> adds) and does not control what a provider does with the data. Those are limits, not details.
